

Fund-Flow Rotation Across Equity Sectors from Public Filings

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Abstract

We measure where investor capital rotates across the eleven sectors of the US equity market, separating investor demand from price performance, using only public data. The input is the net dollar creation and redemption of exchange-traded fund shares, which each fund reports monthly on SEC Form N-PORT. We normalize these flows by fund size, aggregate to sectors, and compare each sector both to the market as a whole and to its own history. The result is a monthly sector-flow panel covering 2019 to 2026 whose reported returns match price-derived returns at a correlation above 0.99, and whose flows and returns reconcile with reported net assets to within about one percent per quarter. On that panel we define the standard rotation statistics: cumulative flow, standardized flow, relative flow, and flow momentum.

1 Flow versus price

A fund's assets under management move for two reasons. The market reprices the holdings the fund already owns, and investors add or withdraw money. Only the second is a flow. Over one period the two combine as

$$A_t = A_{t-1} (1 + r_t) + F_t, \quad (1)$$

where A_t is assets under management at the end of period t , r_t is the fund's total return over the period, and F_t is the net dollar flow. The first term is what the previous period's money grew to on its own. Anything beyond it is new investor money. Rearranging (1) isolates the quantity of interest,

$$F_t = A_t - A_{t-1} (1 + r_t). \quad (2)$$

A flow is a demand signal, not a price signal. Consider a fund worth 100 at the start of a day. The market rises two percent, lifting its holdings to 102 with no investor action. At the close the fund is worth 101. Assets rose, from 100 to 101, yet (2) gives $F = 101 - 102 = -1$: investors withdrew one unit while the price rose. Assets under management and flow can move in opposite directions, which is what makes flow an independent measure of what investors chose to do.

2 Reading the flow

Equation (2) requires assets and return, which still leaves F_t inferred rather than observed. Two routes make it more direct.

2.1 The shares route

An exchange-traded fund issues new shares when demand arrives and redeems them when demand leaves, each at approximately net asset value. The change in shares outstanding therefore reveals the flow,

$$F_t = (S_t - S_{t-1}) \text{NAV}_t, \quad (3)$$

with S_t shares outstanding and NAV_t net asset value per share at the end of t . This is (2) specialized to an ETF: writing $A_t = S_t \text{NAV}_t$ and $r_t = \text{NAV}_t / \text{NAV}_{t-1} - 1$ and substituting recovers (3) exactly.

A share split breaks (3). A two-for-one split doubles S_t overnight with no flow, which the raw difference would read as a large inflow. Let k_t be the split factor on date t , equal to 1 on ordinary days, 2 for a two-for-one split, and $\frac{1}{2}$ for a one-for-two reverse split. The mechanical change is then removed,

$$F_t = (S_t - k_t S_{t-1}) \text{NAV}_t, \quad (4)$$

so a pure split contributes zero flow.

2.2 The reported route

A daily shares-outstanding series for ETFs is not available from free sources. Form N-PORT supplies the flow directly instead. Every US fund files it monthly, reporting for each of the three months in the period the total dollar value of share sales, redemptions, and dividend reinvestment (Item B.8), the monthly total return (Item B.7), and net assets at the period end (Item B.1). The net flow is

$$F_t = \text{sales}_t - \text{redemptions}_t, \quad (5)$$

with reinvestment excluded as internal recycling of distributions. This is the realized data source: monthly granularity, with filings posted about two months after the period. Equation (4) remains the route for daily data when a shares feed is available.

3 Normalizing

A flow of ten million dollars is large for a small fund and negligible for a large one. Dividing by starting size gives the organic growth rate,

$$g_t = \frac{F_t}{A_{t-1}}, \quad (6)$$

read as the fraction of the fund's size that arrived as new money in period t . Dollar flow answers how much capital moved; g_t makes funds of different sizes comparable.

Equation (6) needs A_{t-1} every month, but N-PORT reports net assets only at quarter-ends. We fill the gaps with the identity (1),

$$A_t = \begin{cases} \hat{A}_t & \text{if net assets } \hat{A}_t \text{ are reported in month } t, \\ A_{t-1} (1 + r_t) + F_t & \text{otherwise,} \end{cases} \quad (7)$$

snapping to each reported value and rolling (1) across the two intervening months. Snapping each quarter is required because r_t is a total return: it includes distributions, which leave the fund as cash, so an unbroken roll overstates assets in proportion to dividend yield. Re-anchoring at every report bounds that bias to a single quarter. The remaining sub-quarter error is small enough to leave g_t accurate (Section 6).

4 Aggregating to categories

A single fund is not a sector. We assign each fund i to a category c and sum within it,

$$F_{c,t} = \sum_{i \in c} F_{i,t}, \quad (8)$$

$$A_{c,t} = \sum_{i \in c} A_{i,t}, \quad g_{c,t} = \frac{F_{c,t}}{A_{c,t-1}}. \quad (9)$$

The category growth rate (9) is the asset-weighted average of its member funds' growth rates, so larger funds count more. The market as a whole is the category containing every fund,

$$g_{U,t} = \frac{\sum_i F_{i,t}}{\sum_i A_{i,t-1}}, \quad (10)$$

which serves as the baseline in Section 5.

5 Rotation

Rotation is movement of capital between categories over time. Four statistics describe it.

Monthly flow is noisy. Trailing cumulative flow over a window of W months shows the sustained trend,

$$\text{CF}_{c,t}(W) = \sum_{k=0}^{W-1} F_{c,t-k}. \quad (11)$$

Categories differ in their typical flow volatility, so raw $g_{c,t}$ is not comparable across them. Standardizing against each category's own trailing history of length L gives a comparable score,

$$z_{c,t} = \frac{g_{c,t} - \mu_L(g_c)}{\sigma_L(g_c)}, \quad (12)$$

with μ_L and σ_L the rolling mean and standard deviation. A value $z_{c,t} = 2$ means the category is drawing unusually strong inflows relative to its own norm.

Rotation is inherently relative: capital leaving one category to enter another. Comparing each category to the market separates that movement from a market-wide tide,

$$\text{rel}_{c,t} = g_{c,t} - g_{U,t}. \quad (13)$$

A positive value means the category attracts more than its market-weighted share of flow. Asset-weighting (13) and summing over categories gives zero, since the weighted average of $g_{c,t}$ is $g_{U,t}$ by (10); relative flows are a zero-sum reallocation.

Whether a category's relative strength is rising or fading is its momentum over D periods,

$$m_{c,t} = \text{rel}_{c,t} - \text{rel}_{c,t-D}. \quad (14)$$

Plotting categories by relative flow (13) against momentum (14) places each in one of four quadrants: leading (strong and rising), weakening (strong and falling), lagging (weak and falling), and improving (weak and rising). A category typically traces these clockwise, and the path is the rotation.

6 Validation

The flows are reported rather than estimated, but the reconstructed quantities admit two independent checks. Both run over the eleven-sector panel of Section 7, eighty-one months from July 2019 to March 2026.

First, N-PORT’s reported monthly total return (Item B.7) is compared to the return computed from public prices. Across the eleven sectors the correlation exceeds 0.99, with mean absolute error between 0.17 and 0.80 percentage points. This confirms the return series and the month alignment.

Second, the asset recursion (7) is tested at the quarterly horizon. Starting from each reported net-assets anchor, we roll flows and returns forward to the next anchor and compare the prediction to the reported value. The median absolute error is between 0.2 and 1.0 percent for every sector, confirming that flows, returns, and reported assets are mutually consistent.

A single month of the panel illustrates the output. Table 1 reports net flow and growth rate by sector for March 2026.

Ticker	Sector	Net flow (\$M)	g (%)
XLE	Energy	+2,024	+5.38
XLU	Utilities	+1,044	+4.28
XLY	Consumer Discretionary	+267	+1.18
XLK	Technology	+199	+0.23
XLF	Financials	+80	+0.16
XLV	Health Care	−912	−2.11
XLC	Communication Services	−1,484	−5.46

Table 1: Reported net flow and growth rate by sector, March 2026 (selected rows). Capital enters Energy and Utilities and leaves Communication Services and Health Care.

7 Data and scope

The source is SEC Form N-PORT: monthly, public, with a reporting lag near two months, and coverage from late 2019. The universe is the eleven Select Sector SPDR funds, one per GICS sector, which together tile the US equity market. Each sector currently holds one fund, so (8) reduces to an identity and (9) equals that fund’s growth rate; the aggregation becomes load-bearing once a category holds several funds.

Three data effects are handled explicitly. Splits enter through the factor k_t in (4). Distributions enter through the re-anchoring in (7), which prevents total return from inflating reconstructed assets. The reporting lag means the most recent month is provisional until its filing posts.

ETFs are a large but partial slice of invested capital; mutual funds, separately managed accounts, and direct holdings are outside this universe. The panel is a proxy for sector positioning, not a complete account of it.

A Notation

Symbol	Meaning
t	time period (month)
i	a single fund
c	a category (sector)
U	the whole-market aggregate
S_t	shares outstanding at end of t
NAV_t	net asset value per share at end of t
A_t	assets under management, $S_t \text{NAV}_t$
r_t	total return over t
k_t	split factor (1 ordinarily, 2 for two-for-one)
F_t	net dollar flow over t
g_t	normalized flow, F_t/A_{t-1}
$\text{CF}_{c,t}(W)$	trailing cumulative flow over W months
$z_{c,t}$	standardized flow over lookback L
$\text{rel}_{c,t}$	relative flow, $g_{c,t} - g_{U,t}$
$m_{c,t}$	flow momentum over D periods